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Understanding the development of flood management in the middle Yangtze River

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ABSTRACT

This study aims to investigate whether, to what extent and how a transition toward integrated flood management has taken place in the Dongting Lake area at the middle Yangtze. Accordingly, we conducted a longitudinal research of its flood management (1949–2009). We developed an analytical framework linking regime components to two societal learning types (double and triple-loop learning) that are key to a regime transition. Our study shows that the transition toward integrated flood management has already started, but the whole regime transformation will still take time to complete, due to, for example, the not-yet-ready decision-making processes that shape the structure changes as well as the incompatibilities between what is on paper and real implementation. To understand how the regime transition took place, we investigated where and how triple-loop learning was initiated as well as how so-called "informal learning processes" has contributed to the transition of Dongting flood management.

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1. Introduction

Flood control has long been an engineering-centered approach that aimed at lowering the probability of flooding and reducing society's exposure to floods, within a paradigm of "absolute protection" (Schanze, 2006). By realizing the complex nature of flood disasters and impacts of flood control measures, the international community has increasingly committed itself to an integrated approach for flood management to address risk management as well as the economic, social, and environmental effects of different measures (Schanze, 2006; APFM, 2009).

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The Yangtze, the longest river in China, has been subjected to flooding throughout history. Especially the development of the provinces and municipalities at its middle and lower reaches, which are of great economic importance for China, have been hampered by disastrous floods. Thus, dealing with floods has been the major concern of not only these local riparian governments but also the central government. Following the disastrous floods in the Yangtze and other major rivers in 1990s, the Chinese government has also realized that engineering-centered flood control alone is not sufficient to deal with the complex flooding system and expressed the need of a paradigm shift from traditional flood control to an integrated approach of flood management (E, 2004).

Extensive research has contributed to the knowledge base of integrated flood management in China and the Yangtze River Basin, ranging from delivering a theoretical framework of flood management strategies (e.g. Cheng, 2004) and innovative policy instruments (e.g. Xiang, 2002) to flood risk management (e.g. Gemmer, 2004). However, the developments that have really taken place in flood management have received far less attention (Te Boekhorst et al., 2010). Given the persistent nature of water problems (Van der Brugge and Rotmans, 2007), the investigation of the flood management development should take a system perspective, i.e. the structural change of a regime, including its management paradigm, policies, physical infrastructures, multi-actor decision-making processes, etc. Pahl-Wostl (2007) claims that improving the understanding of such a development is essential for supporting the transition toward a desired regime, e.g. integrated flood management. A vast array of studies have provided theoretical bases and heuristic frameworks for studying transition (e.g. Geels, 2002; Elzen et al., 2004; Loorbach, 2007; Rotmans, 2005). However, transition research has focused largely on socio-technological systems in the context of developed countries where transitions are often driven by technical innovations. Transition in societal systems that fulfills specific functions, such as water management, or transition in the context of developing countries, such as in China, may involve different kinds of dynamics and be driven by other key factors and deserves in-depth study. Thus, investigating the development process of a societal system in China will broaden the perspective of the current transition research.

Accordingly, this study aims to investigate whether, to what extent and how the transition toward a regime of integrated flood management (IFM) in the middle region of the Yangtze River has taken place. To this end, we conducted a longitudinal study (1949–2009) of flood management for the Dongting Lake area, which is mainly located in Hunan Province at the Middle Yangtze. Dongting Lake is a retention lake naturally connecting to the Yangtze River and is very important for storing excessive flood water from the main channel of the Yangtze, and has thus been designated by the central government as a key national Flood Retention and Storage Area (FRSA). On the other hand, the lake is also key for the socio-economic and cultural development in its surrounding area, due to its rich natural assets. However, its development has also long suffered from disastrous floods (Lai and Mo, 2004).

2. Analyzing the transition of a flood management regime

Prior to exploring a regime transition, we first present what is meant by the term "regime" in this paper. No single standard definition of "regime" is widely accepted (Holtz et al., 2008). Generally, the notion of regime can be distinguished between two schools. The first conceptualizes regime as a socio-technological system that centres on technologies. The second elaborates regimes tailored to a sub-societal system that fulfils specific functions, such as water management (Van der Brugge, 2009; Holtz et al., 2008). The "regime" in this paper follows the later conceptualization. We applied the regime concept described by Van der Brugge (2009) to support the empirical analysis of a regime development by unfolding regime components and their internal dynamics. He characterizes a regime as a system containing interactions of three components: actor,[1] process,[2] and regime structure and further distinguishes three interconnected *components in the regime structure*:

[1] According to Van der Brugge (2009), the "actor" component includes both individuals and organizations.

[2] The "process" component connects "the sphere of actors to the sphere of structures" of the regime (Van der Brugge, 2009). It can include management practices such as strategic planning as well as initiatives outside the government agencies.

- the *culture structure*, which includes a management paradigm, discourse, values and knowledge base;
- the *institution structure*, i.e. regulative institutions;
- physical infrastructures.

The structure emerges and evolves as a result of the processes, where concerned actors are engaged in (Van der Brugge, 2009).

Among these three structural components, in the scientific literature, it is claimed that the paradigm in the culture structure shapes the regime development and has the power to exclude non-compatible approaches (e.g. Dosi, 1982; Pahl-Wostl, 2007; Pahl-Wostl et al., 2011). A lack of reflection on the paradigm of an established regime can severely constrain the regime transition (Pahl-Wostl, 2009). Pahl-Wostl (2009) thus conceptualizes the change of a natural resource management regime as societal learning and developed an analytical framework based on the concept of "single-, double-, triple-loop learning". Single-loop learning refers to an incremental improvement of established routines without questioning the underlying management paradigm. In contrast, double-loop learning calls into question the underlying management paradigm. Triple-loop learning implies the dominance of a new management paradigm and a corresponding transformation in the rest of the regime shaped by the paradigm. Regime change is assumed to proceed in a stepwise fashion moving through the phases of single to double on to triple-loop learning.

To judge whether a transition in a given natural resource management regime in fact takes place, and to what extent, we developed an analytical approach that links these three learning types and the expected features of the three structure components of a regime under the Integrated Flood Management (IFM), as shown in Table 1. In the light shaded part, indicators of different levels of learning are listed. Transition requires questioning the paradigm of the established regime (Pahl-Wostl, 2009). Thus, we claim that a regime starts to transition when double-loop learning regarding the management paradigm toward IFM takes place. This should be reflected in the cultural component where the paradigm is questioned beyond isolated groups. Furthermore, mainstream actors start experiments in the institutional and infrastructure components of the regime that reflect new management paradigm. We consider a regime to be transformed when triple-loop learning is prevalent in all structure components. In the dark shaded part, we specified the major normative features of the three structural components of the regime under IFM, based on a literature review of flood management and integrated and adaptive water management (e.g. APFM, 2009; GFDDR, 2012; Pahl-Wostl, 2007). Combining both light and dark shaded parts enables us to identify which level of learning toward IFM takes place in which structural components of the flood management regime.

Meanwhile, triple-loop learning toward IFM in all structural components requires changes in decision-making or implementation processes that shape the structure changes (we listed three major indicators of IFM in the last part of Table 1). We asked whether these processes are in place to support the transformation to IFM.

To understand how transition of the flood management regime has taken place, we focused on the following two aspects. Firstly, we analyzed at which administrative level triple-loop learning in the regime structure was initiated and how. Second, double-loop learning requires settings supporting informal[3] social learning, where established paradigm is questioned, novel measures are experimented (Xia and Pahl-Wostl, 2012a). Thus, we attempted to find out whether such informal learning processes exist and to understand how they were linked to formal policy processes[4] that guided Dongting Lake flood management.

[3] The informal nature is manifested in the following two aspects: (1) the results of these processes are not legally binding and do not result in major changes in regime structure; (2) the participants – both governmental and non-governmental actors – join the learning processes either inside or outside their obligations (Xia and Pahl-Wostl, 2012a).

[4] The formal policy process includes activities ranging from state/policy assessment, goal setting, policy formulation and policy implementation to monitoring & evaluation (Pahl-Wostl, 2007).

Table 1
different levels of learning and features of flood management regimes.

Regime components		Double-loop learning	Triple-loop learning
Regime structure	**Culture**	● The existing paradigm of exclusive flood control is questioned beyond isolated group ● Ideas and knowledge reflecting the IFM paradigm emerge beyond isolated groups	● The IFM management paradigm and corresponding discourses becomes dominant ● Powerful "mainstream" actors promote the IFM paradigm
		IFM paradigm: *perceiving that the flood risk can be reduced and floods can also benefit societal development and ecosystems; living with floods*	
	Institutions and infrastructures	Mainstream actors start experiments in the institutional and infrastructure components of the regime that reflect new management paradigm of IFM	● Formulation of regulation framework and wide enforcement of institutional measures of IFM below ● Wide implementation of infrastructure measures of IFM below
		IFM Institutional and infrastructure measures: ● *Adjustment of land use and development on floodplains to reduce society's exposure to flooding* ● *Society's preparedness for floods is enhanced and its vulnerability reduced by various non-structural measures* ● *Social equity is promoted via cost and benefit sharing mechanisms among various stakeholders* ● *Measures that take advantage of the flood regulation service of a floodplain are explored* ● *The adverse impacts of flood control structural measures on ecosystems are taken into account* ● *an integrated approach that deploys not only structural but also non-structural measures, taking into account the flow exchange (water, sand, pollution) between the lake and the river*	
Actors and process		New decision-making and implementation processes in line with IFM emerge	Decision-making and implementation processes in line with IFM become dominant
		Decision-making and implementation processes under IFM: ● *Cross-sectoral analysis identifies emergent problems and integrates policy implementation* ● *Participatory processes, including non-governmental stakeholders and experts from other disciplines* ● *An agency, who are authorized to coordinate the interests of different riparian provinces of the whole river basin, to enforce key flood management measures, and to avoid flood risk transfer between up- and down-stream, should be in place.*	

To operationalize the analytical framework, we first identified critical events, where policies that explicitly guided Dongting Lake flood management (e.g. strategic plan for institutional and infrastructure improvement in the area) at national, river basin, and provincial level were formulated, and then searched for informal learning processes closely linked to these events or key general flood management policies. We applied Management and Transition Framework (MTF), an analytical tool, to structure these events and their linkages during the development of a regime. The core of the MTF is the "Action Situation (AS)", which refers to "a structured social interaction context that leads to specific outcomes". This means each critical event and informal learning process identified can be regarded as an AS. Each AS has a series of input and output features, such as the "institution", "operational

outcome", "actors",[5] and so on (Pahl-Wostl et al., 2010). These features are very useful in characterizing a regime, i.e. its structure as well as its actors and processes. Accordingly, the MTF tool can help to:

- examine whether and to what extent the flood management regime has developed toward a regime of IFM by comparing the "institution" and "operational outcome" feature of ASs with the normative features of IFM regime structure;
- find out at which administrative level (national, river basin, and provincial, etc.) triple-loop learning in the regime was initiated by assessing which "actors" lead the ASs containing triple-loop learning;
- understand how informal learning processes were linked to formal policy processes by zooming into the "operational outcome" and "actor" feature of the identified informal learning ASs.

This study combined desk research with expert interviews for data collection. Desk research included sources from government websites, government reports, NGO reports and scientific literature. Interviewees included policy-makers from water management agencies and experts from the design and planning institutes of water management agencies, WWF staff and scientists familiar with the development process of flood management of the Dongting Lake area.

3. Development of the flood management regime in the Dongting Lake area between 1949[6] and 2009

3.1. Regime structure development

Since the founding of the PR China in 1949, flood management pursued a combined approach of land reclamation for agriculture and the creation of Flood Retention and Storage Areas (FRSA): during the non-flood season, gates were closed to prevent the river from flowing into FRSAs, supporting agricultural production in those areas; during the flood period, the gates were opened to give the flood space (WWF, 2003). In reality, land reclamation was intensified, due to the primary national development goal of increasing grain production by all possible means[7] and the good natural conditions for agricultural production in the Dongting Lake area. Land reclamation was followed by settlement (WWF, 2003). Consequently, land reclamation with dike construction became the predominant practice. As a result, the surface area and water storage capacity of the lake decreased dramatically. The size of Dongting Lake declined from 4350 km^2 to 2820 km^2 between 1949 and 1971 (Yang et al., 2007).

Against this background, the Ministry of Water Resource and Hydropower (MWRH)[8] organized a series of symposiums about flood control at the middle and lower Yangtze with representatives from riparian provinces of the Yangtze between 1969 and 1980 (Zhong, 2007) [AS1 and AS2]. The symposiums delivered the following outcomes: (1) designation of the Dongting Lake area as a place to store 16 billion m^3 flood water when facing floods of the same scale as the 1954 flood[9]; (2) ban on land reclamation at the Middle Yangtze, and (3) specification of the major objectives of flood management in the short term—"Dike reinforcement and FRSA construction". Large-scale land reclamation ceased in the Dongting Lake area in the 1970s. Consequently, the size of the Dongting Lake decreased by only 4%[10] between 1971 and 1983, compared to 35% between 1949 and 1971 (Yang et al., 2007) (Table 2).

[5] The "institution" here refers to **regulations, strategic plans, paradigm, visions**, as outcomes of correponding ASs. The "operational outcome" feature represents the **implementation** of strategic plans, policies, infrastructural measures, and programs. The "actors" can help to understand who are participated in and leads a specific process, such as strategic planning, pilot project, etc..

[6] Given the data constraints, the AS identification starts from 1969.

[7] Land reclamation was one of the key measures guided by the paradigm of "Mankind Must Conquer Nature", insisted upon by Mao (Shapiro, 2001).

[8] The current Ministry of Water Resource (MWR).

[9] The 1954 flood was the biggest flood in the Yangtze River basin in the 20th century.

[10] The decrease in size could also have been caused by natural processes such as silt deposition.

Table 2
Action situations (ASs) 1969–1995.

No.	Year	ASs	"Institution" class	"Operational outcome" class
1	1969	Symposium 1969[a]	Designation of the Dongting Lake area for flood storage	[c]
2	1980	Symposium 1980[a]	(a) Stop further land reclamation; (b) specify near-term flood management objectives	Large-scale land reclamation stopped.
3	1986–1995	Formulation and implementation of "*Near-term plan*" (1st period project)	(a) Dike reinforcement for key polders; (b) safety construction of FRSAs; (c) expansion of the flood discharge channel; (d) improvement of the infrastructures of flood communication; (e) polder categorization[b]	(a); (b); (d); and technical experiment on channel regulation
4	1991–1993	Hunan's pilot program on voluntary flood insurance		Implementation of a pilot program; demonstrating infeasibility of voluntary insurance
5	1993–1995	Formulation of "*Near-term plan*" (2nd period project)	(a) Further reinforcement of dikes of key polders; (b) channel regulation; (c) further safety construction of FRSAs; (d) further improvement of flood communication infrastructures	(a); (b) *(implementation at phase 2)*

[a] Dongtingwang (2010).
[b] Polders were categorized according to key polders where its settlement must be protected and a dike should be reinforced to national standard, polders with lower protection priority, and FRSA polders
[c] Empty cells in the tables mean that no data available.

In order to store 16 billion m^3 of flood water and to achieve the near-term major flood management objectives specified in the symposiums, ***Hunan Water Resource Department (WRD)*** formulated two plans for Dongting flood control [AS3] and [AS5], approved by the central government (HHPDI, 2009).

In 1988, the State Council issued the "*Guidance of safety and construction of FRSA*", which was formulated by Ministry of Water Resources (MWR). For the first time the "*Guidance*" specified management measures that included flood insurance or funds, population control and land use constraints. In response, the ***Hunan provincial government*** required its WRD to support the state-owned China People Insurance Company to implement a pilot program on voluntary flood insurance in villages in high-risk FRSAs between 1991 and 1993 [AS4]. The outcome demonstrated the infeasibility of voluntary flood insurance due to the low incentives of both the affected communities and the insurance company. Consequently, further expansion within the province has been stagnant. Also, feedback from the results of experiments conducted in Hunan and the other provinces triggered the consideration of mandatory flood insurance at the national level (HHPDI,[11] personal communication, May 5, 2009).

Directly after the flood at the middle reach of the Yangtze in 1996, a symposium of "Studies on integrated Dongting Lake management" was held by ***the State Council*** [AS7]. It proposed an integrated engineering approach toward flood management, taking into account diverse measures at different parts of the Dongting Lake area[12] and the flow exchange (water, sand, pollution) between the lake and the Yangtze River (Guo, 2008). This strategic change influenced the formulation of the "*Near-term plan of integrated management of the Dongting Lake area*" developed by the ***Changjiang Water Resource Commission (CWRC)***[13] ***and riparian provinces*** [AS6]. The "*Plan*" proposed structural measures to be complemented by non-structural measures such as flood communication, flood insurance piloting, and FRSA regulations (HHPDI, 2009).

[11] HHPD refers to the Hunan Hydro & Power Design Institute. It has played a major role in drafting strategic plans for Dongting Lake.
[12] Water soil conservation at the upper stream; building up reservoirs at the middle reach; dredging and dike reinforcement at the lower part.
[13] In China, river basin commissions are subordinate agencies of Ministry of Water Resource.

In 1998, the river basin-wide flood affected 223 million people and caused direct economic losses amounting to 166.6 billion Yuan. During the flood period, both the flow at flood peak and the total volume of flood were less than those experienced in the 1954 flood, but water levels in rivers and lakes were higher (Zhong and Chen, 2000). It revealed the increasing tension between human development and the flood regulation service of the floodplain, which received considerable attention from the central government. Meanwhile, the floodplain management approach employed by the US government after the Mississippi flood in 1993 inspired the central government to consider floodplain restoration (IWHR, personal communication, Apr 17, 2009). As a result, the ***State Council*** issued the "*32 Characters policy*", which requested "*Demolishing polders and dikes for flood discharge, Restoring reclaimed farm lands into lakes, and Relocating residents to build up new townships (termed as DRR)*" [AS8]. The ***State Planning Commission***[14] ***(SPC) led six ministries*** to formulate the "*Report about the detailed implementation of DRR in Hubei, Hunan, Jiangxi, and Anhui Province*", aiming at restoring 2900 km^2 with 3.36 million people to be relocated in these four provinces with a budget of 10.1 billion Yuan [AS9] (Gao, 2002). Accordingly, ***Hunan provincial government*** established DRR offices at different administrative levels. The Provincial DRR office was responsible for drafting a DRR action plan over the next four years (1999–2002) and for supervising its entire implementation [AS10]. To facilitate the implementation, the provincial government also issued different policies, such as subsidies, tax reductions and exemptions, land use, etc. The goal of the DRR action plan was achieved by 2005. In 2002, during his field trip to Hunan, ***the former Premier Zhu Rongji*** strengthened the importance of a long-term plan for the development of the Dongting Lake area and suggested that the first step should be to restore the Dongting Lake to 4350 km^2, which was the size of the lake before 1949 [AS13]. In response, ***the Hunan provincial government*** established the provincial "4350 office" and drafted the ambitious "*4350 Plan*" in three months [AS14]. However, this plan was never implemented, due to the retirement of Premier Zhu and the subsequent insufficient financial support from the central government as well as the huge difficulties involved in relocating people (HHPDI, personal communication, May 5, 2009). Meanwhile, MWR issued the "*Opinions about near-term flood control in the Yangtze River Basin*" in 1999, which specified that priority should be given to the construction of FRSAs that can hold 5 billion m^3 in Dongting Lake area in Hunan Province [AS12] (Table 3).

On the other hand, following the implementation of the "32 Character" policy, the local government did not fully consider the communities' livelihood after relocating people. The international NGO—***WWF***—realized that this would probably undermine the achievement of lake restoration and started its program of "Partnership for a Living Yangtze" and pilot for supporting livelihood in polder areas. During the program, WWF collaborated with research institutes, provincial forest department, county government, and local community. The first program phase evaluated the feasibility of polder economy at the middle reaches of the Yangtze. The results were disseminated to the governments at different levels. Additionally, the program attempted to explore and demonstrate the "flood adaptation economy for FRSA communities" as an alternative of communities' livelihood strategy (WWF, personal communication, April 29, 2009) [AS11].

In 2000, the State Council issued the "*Interim regulation of compensation for the use of FRSAs*", which specified the mechanism of providing compensation to residents in FRSAs for the use of FRSAs during major floods. After a large flood in 2003, ***provincial government*** compensation to households in its Li Nan polder[15] was implemented, which was regarded as a success by the central government [AS15] (Hunan Daily, 2003).

In 2003, at the national symposium on alleviating flood and fighting drought, the Minister for the Ministry of Water Resource (MWR) emphasized "from flood control to (integrated) flood management". This constitutes a distinctive holistic perspective for dealing with floods. It implies the following new discourses: (1) "living with floods"; (2) "flood risks should be shared among the whole of society"; (3) "consideration of the negative impacts of structural measures", and (4) "enhancing society's preparedness for flooding" (E, 2004). These new principles have become the written guiding

[14] The current National Development and Reform Committee.

[15] Li Nan is a national designated key FRSA. DRR policy had already been implemented in this polder before 2003, meaning that the compensation burden for the government was insignificant.

Table 3
Action situations (ASs) 1996–2009.

No	Year	AS	"Institution" class	"Operational outcome" class
6	1996[a]–1998	Incorporation of the "integrated approach" in the formulation of the "*Near-term plan of integrated management*"	Specification of integrated flood control, in terms of employing structural measures complemented by non-structural measures	Mainly the implementation of the "institution" class of [AS6]
7	1996	Symposium on studies on integrated Dongting Lake management	Proposal for an integrated engineering approach toward flood control	
8	1998	"32 Character" policy issued	Floodplain restoration	
9	1998	Formulation of the DRR Plan in the Yangtze River Basin	Measures and funding arrangement for DRR were specified	
10	1998–2005	Formulation of the provincial DRR Action Plan	Aimed to destroy 333 polders and to relocate 558,000 people	Completely implemented
11	1998–2008	WWF program "Partnership for a Living River"		Demonstrated flood adaptation economy for FRSA communities
12	1999	Issuing "Opinions about near-term flood control in the Yangtze River Basin"	Prioritized the construction FRSAs in Dongting Lake area	
13	2002	Premier Zhu visited the region	Suggestion to restore Dongting Lake to 4350 km^2	
14	2002	"4350" project	Establishment of the "4350" office and draft of the ambitious "4350" Plan	Not implemented
15	2003	Provincial implementation of compensation		Successful compensation to households in Li Nan polder
16	2005–2009	Revision of the "*Near-term plan of integrated management*"	• Multiple objectives under the concept of "integration" • More diverse non-structural measures • Categorization of FRSAs	Partly implemented
17	2008	Piloting FRSA construction and management		The dikes in the safety area were strengthened and the feasibility of relocating people to establish new townships was explored

[a] Although the development of the Plan commenced in 1994, the "integrated approach" idea was expected to be included in the planning from 1996.

principle in policy-making of flood management at different administrative levels, including Dongting lake flood management.

In 2006, MWR together with the National Development and Reform Commission (NRDC)[16] and Ministry of Finance (MOF), proposed the "*Opinions on strengthening the safety construction and management of FRSAs*". The "*Opinions*" prescribed a general direction of FRSA construction and management, including: (1) safety area construction; (2) FRSA management, such as drafting flood risk maps, regulating the socio-economic development and land use, and drafting FRSA plans; and (3) improving the compensation mechanism. ***MWR and NDRC*** then called for experiments on FRSA construction and management to be conducted in three key FRSAs of the Dongting Lake area [AS17]. The experiment conducted by the ***Hunan provincial government*** aimed to explore the feasibility of relocating people to a new township and of constructing safety infrastructures. The planned measures included: (1)

[16] NDRC has a function to co-ordinate actions among different ministries.

dike reinforcement of safety areas; (2) planning the new township in accordance with the "*construction of small towns*" principle[17]; (3) construction of safety areas; and (4) construction of roads and communication infrastructures (HNWR, 2005).

Given certain new conditions, such as the operation of the Three Gorges Dam and various institutional changes at national level, ***CWRC*** led the revision of "integrated management plan of Dongting Lake area" [AS16] (HHPDI, 2009). "Integration" in the plan was given a broader meaning in the revised plan: (1) except for flood management, other aspects of water resource management, such as water quality management were also acknowledged; (2) it aimed at integrating water management activities with policies of "new rural China construction"; (3) it addressed land use and the development of FRSAs. Non-structural measures in the revised plan were also more diverse than those in the first one. The following measures were proposed: (1) the study of flood insurance mechanisms; (2) the formulation of policies and regulations for socio-economic development in FRSAs; (3) investigation of compensation mechanisms and the drafting of relevant regulations. Also, FRSAs were categorized in accordance with their probabilities of use and importance, to guide the construction, management, and development of FRSAs.

Currently, FRSA management and construction still faces numerous challenges. First of all, FRSA management still focuses largely on structural measures, which is mainly attributed to: (1) a lack of financial resources and (2) the existing ageing and low-quality flood defense system due to the stagnancy of hydraulic infrastructure construction in the 1980s (WRD, personal communication, May 8, 2009). Despite the focus on structural measures, FRSA infrastructures remain uncompleted (e.g. gates to allow flood water to enter and exit), which has limited the role of FRSAs in both flood regulation and supporting the residents' livelihoods. This has mainly been caused by the shortage of financial resources. Secondly, to implement non-structural measures, regulatory frameworks, such as those regulating compensation and flood insurance, were still lacking at the provincial level. Thirdly, rapid development in some key FRSA has limited the use of FRSAs and may cause tremendous socio-economic losses during flooding (Deng et al., 2009). Finally, the accountability issue and the means of implementation were not clearly specified in current policies guiding the FRSA management.

3.2. Strategic planning

Fig. 1 shows the processes in formulating the strategic planning for the latest "plan of integrated management in the Dongting Lake area". Although the Dongting Lake is mostly located in Hunan province, due to its importance for Yangtze flood management, Changjiang Water Resource Committee (CWRC) has also been engaged in the strategic planning process. More recently, CWRC has taken the leading role of organizing the planning process and determining key elements of the Dongting plan.[18] The provincial WRD drafted the plans accordingly. On the other hand, it is provincial and municipal governments who have the predominant power of decision on how to implement the river basin strategic plans (DLWRAB,[19] personal communication, May 8, 2009).

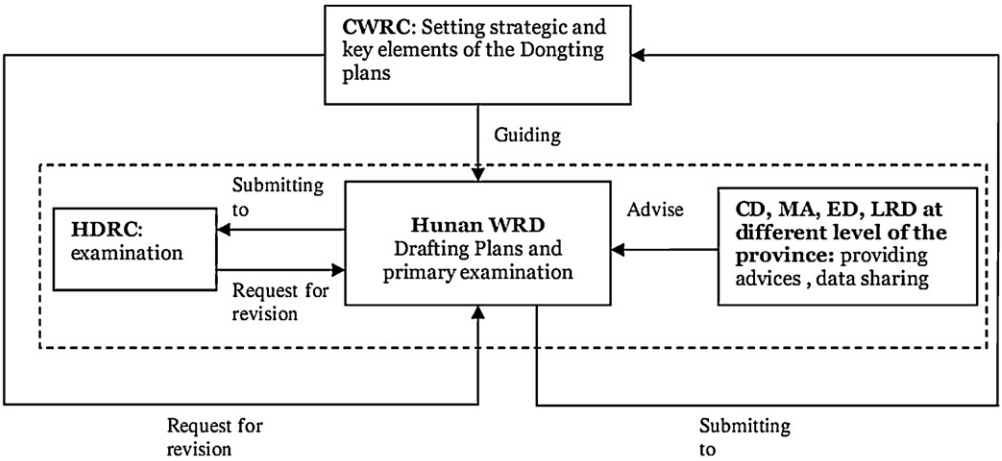
Moreover, Fig. 1 also demonstrates that actors from different departments at provincial level were involved by providing advice and sharing information. However, advice seeking from other departments was often pro forma. For example, WRD was asked by LRD to give advices on the land use plan related to water management purposes. Interestingly, WRD perceived it as a pro forma as always, thus, they had not devoted much effort to it and knew afterwards that LRD really wanted to seek advice from them (HHPDI, personal communication, May 5, 2009). However, during the DDR planning and implementation led by the provincial government, cross-sectoral collaboration among relevant departments did take place. The DRR office was led by the Secretary of the provincial Communist Party Committee[20] and consisted of representatives from all relevant departments (HHPDI, personal communication, May 5, 2009).

[17] The "construction of small towns" aims to urbanize rural China, in terms of its scale, economic structure, and life style, etc., by building up both institutions and infrastructures.

[18] In earlier times, provincial WRD was the leading actor in planning.

[19] Dongting Lake Water Resource Administration Bureau of Hunan Province.

[20] The provincial Committee defines tasks and sets the priorities for the provincial government.



WRD: Water Resource Department; CD: Communication Department; MA: Meteorological Administration; ED: Environmental Department; LRD: Land and Resource Department; HDRC: Development and Reform Commission

Fig. 1. Processes in formulating the latest short-term plan for integrated management in the Dongting Lake area.

In addition, hydraulics and hydrology experts working for the research institutes affiliated to or closely linked with water management agencies have also contributed to drafting the strategic plan. Except for them, no other non-governmental actors were involved in planning (HHPDI, personal communication, May 5, 2009).

4. Discussion

4.1. Has the transition toward IFM taken place?

In Table 4, we examine whether major developments in the three regime structural components reflect the features of IFM and to what extent a transition has taken place, i.e. double-loop or triple-loop learning, based on Table 1. The analysis indicates, except for considering adverse impacts of flood control structural measures on ecosystems, triple-loop learning of the IFM paradigm in the cultural component as well as double-or triple-loop learning of almost all features of IFM in the institutional and infrastructure structure have taken place. Thus, we claim that transition toward the IFM regime has started in Dongting Lake area. It is worth noting that triple-loop learning of floodplain restoration after the flood in 1998 was rapid. Our results also show that triple-loop learning in the cultural structure took place after the triple-loop learning in the other two structures.

However, triple-loop learning toward IFM has been constrained by the incompatibilities between what is on paper (i.e. announced changes in management paradigm, formulated institutions, and planned infrastructures) and real implementation. One example is FRSA construction and management. The continuous focus on structural measures is mainly attributed to an insufficient reconfiguration within the institutional structure and a lack of financial resources. The rapid socio-economic development in some key FRSAs is caused by local governments' interest in economic growth, which is among the most important indicators evaluating local governors' performance. Besides, since FRSA management is no longer solely shaped by the flood management regime, a lack of reconfiguration in other management regimes such as rural community development also hinders the development in key FRSAs. Another incompatibility probably exists between the announced paradigm shift at national level and its implementation. Shen (2010) depicts the implementation of new paradigm as a "Herculean task", due to a long and strong belief in engineered measures among

Table 4
Learning toward IFM in Dongting Lake area flood management.

Structural components	Features of IFM	Findings: development of the regime structure	Level of learning
Culture	Perceiving that the flood risk can be reduced and that floods can also benefit societal development and ecosystems; living with floods	***[N]2003: The central government explicitly announced that national flood management should shift toward integrated flood management, which has also served as the principle for Dongting flood management and has been integrated into local policy-making***	T–C
Institution and infrastructure	Adjustment of land use and development on floodplains to reduce society's exposure to flooding	***[N]1969 and 1980: ban on land reclamation*** [N] + [P]2008: the central government called for experiment of FRSA construction and management, which was implemented by Hunan government	D–(I,P) + T–I
	Enhancing society's preparedness for floods and reducing its vulnerability	[P]1991–1993: flood insurance experiment, which was encouraged by the national "Guidance" [RB]2005–2009: further study on flood insurance mechanisms [N] + [P]2008: experiment of FRSA construction and management	D–I
	Social equity is promoted via cost and benefit sharing mechanisms among various stakeholders	[P]2003: small-scale implementation of compensation for the use of FSRA, which was backed up by national "*Interim regulation*" [RB]2005–2009: further study on compensation mechanisms and plan to draft relevant regulations	D–I
	Exploring measures that take advantage of the flood regulation service of a floodplain	***[N]1969 and 1980: The central government designated Dongting Lake area for flood water storage.*** ***[N]1998: The central government ordered rapid implementation of floodplain restoration and set up targets and corresponding organizational structure.*** ***[P]1998–2005: The provincial government effectively implemented restoration with different policy measures***	T–(I,P)
	The adverse impacts of flood control structural measures on ecosystems are taken into account		
	Integrated approach	[N]1996: an integrated engineering approach toward flood control was proposed in the national symposium. ***[RB]1996–1998: The "near-term plan" prescribed that structural measures should be complemented by non-structural measures*** ***[RB]2005–2009: The "plan of integrated management of the Dongting Lake area" gave broad meaning to "integration" which includes consideration on land use and socio-economic development in FRSA and* further encourages non-structural measures.** [N] + [P]2008: experiment of FRSA management	D–(I,C) + T–I

[] represents the administrative level of learning: N, National; RB, River Basin; P, province; "D", double-loop; "T", triple-loop (bold and italic text); "I", institutions; "P", physical infrastructures; "C", culture.

water managers in China. Deeply entrenched beliefs are not simply changed by a top-down push. As soon as these incompatibilities exist, transition can get stuck or the regime may even reverse to the original status (Van der Brugge, 2009; Pahl-Wostl et al., 2011).

4.2. Are decision-making and implementation processes supportive of the transformation to IFM?

We found evidence of cross-sectoral collaborations in ASs initiated at national level that supported double- and triple-loop learning after the disastrous flood, for example; six ministries' collaboration

on formulating the DRR plan led by the State Plan Commission (SPC) [AS9] and the call for FRSA management experiments initiated by MWR and NDRC [AS17].

However, cross-sectoral collaboration at provincial level still appears to be insufficient. Water management agencies have always been the decisive actors in the strategic planning process; while other departments have mostly played a passive role. Since water management agencies are staffed exclusively with water professionals, it is obvious that the strategic decisions of flood management are made with little involvement of other different disciplines. Although floods have long been effectively managed by the water management agencies in a top-down fashion, a lack of an active cross-sectoral and interdisciplinary analysis can be a key obstacle for triple-loop learning of, for example, adjusting land use or socio-economic development in the FRSA. Signals for cross-sectoral collaboration were identified in recent practices. However, it can be expected that a change of actors' mind sets for such collaboration may take time, due to the lack of co-operational experiences in the past. In fact, cross-sectoral collaboration in policy formulation and enforcement can be quite effective when the provincial government leads the process. Beyond the Dongting case, the provincial government also plays a powerful role in facilitating cross-department data sharing and assessment. If a water-related project are funded by the central government, which requires comprehensive assessments, the provincial government will take the co-ordinator role and require all relevant departments to provide data listed by WRD (HHPDI, personal communication, May 5, 2009). In addition, there is a general lack of non-governmental actors participating in strategic planning of Dongting flood management.

Furthermore, although there is an increasing role of river basin commission, CWRC, in strategic planning, provincial and municipal governments have the predominant power of decision over its implementation. Given the economic development as the priority of the local governments, a lack of power by the river basin commission over the enforcement can be another factor for the incompatibilities discussed above.

To sum up, the decision-making and implementation processes of Dongting flood management have not yet been ready for supporting triple-loop learning and the transformation to IFM.

5. How has the transition toward a regime of IFM taken place?

Our study demonstrates that triple-loop learning toward IFM in the Dongting Lake area has been mostly initiated at national level (Table 4).[21] Such a top-down driven pattern as in our case study can be distinguished in the following three ways. First, the central government issued new policies that are in line with IFM. For example, the "32 Characters" policy [AS9] specified clear targets and ordered immediate implementation. Second, as soon as a new discourse (e.g. the announced new paradigm) was announced by the central government, it will directly be included in discourses at different administrative levels. Third, the attention from a high-ranking politician also led to triple-loop learning in institutional and infrastructures of the flood management regime. Former Premier Zhu's speech immediately drove the formulation of the ambitious action plan for lake restoration and the establishment of a corresponding organizational structure, without a participatory-based integrated assessment of the feasibilities of such a large-scale restoration. This kind of triple-loop learning is difficult to sustain as soon as the attention is shifted away or the politician leaves his/her position.

On the other hand, three informal learning processes were identified, which can be distinguished between two types. Table 5 depicts these learning processes linked to an AS in the formal policy process. The first type is government-initiated policy experiments ([AS4] and [AS17]). These experiments generally aim to examine the feasibility of certain institutional and infrastructural measure(s) and deliver lessons for wide implementation if they are successfully implemented. This type is initiated by the government and thus has stronger links to the formal policy process, i.e. the actors are within the formal processes.

[21] Although triple-loop learning was initiated at river basin level in two ASs , it was regarded to be initiated by the central government, given the fact that river basin commissions are subordinate agencies of the Ministry of Water Resource in China.

Table 5
Examples of informal learning processes and their links to formal process and contribution to learning.

Informal learning example	Actors		To which critical events or key policies it was linked?	Contribution to double-or triple-loop learning
	Actor of AS	Their links with formal policy process		
Flood insurance piloting in Hunan [AS4]	• Provincial government and State-owned China People Insurance Company (leading) • Hunan WRD	Within the formal policy process	The State Council issued the "Guidance of safety and construction of FRSA" that encouraged piloting on flood insurance.	They are themselves double-loop learning on innovative institutional and infrastructure solutions
FRSA management experiment [AS17]	MWR, NDRC, Hunan province	Within the formal policy process	MWR and NDRC called FRSA experiment in response to the issuing of "Opinions on strengthening the safety construction and management of FRSAs".	
The "Partnership for a Living Yangtze" program [AS11]	• WWF (leading) • Research institutes • Provincial forest department • County government • Local community	No	It sustained the results of the successful implementation of "32 Characters" policy of flood restoration [AS10] in the pilot areas.	It sustained the rapid triple-loop learning of institutions and infrastructures for floodplain restoration by governments at different administrative levels.

The second type is a pilot initiated by non-governmental actors [AS11]. It did not have actor linkages to the formal process of flood management. Water management agencies did not join the pilot project due to its focus on wetland conservation. However, beyond the Dongting case, WWFs networking with MWR and CWRC through the later Integrated River Basin Management (IRBM) Task Force initiated by the CCICED[22] has delivered their pilot lessons and their ecological approach to the formal policy process of flood management at the national and river basin level (Te Boekhorst et al., 2010). This could potentially shape flood management at the provincial level, given the hierarchical nature of Chinese water management. Besides, WWFs close collaboration with the media enables the informal learning outcomes to reach policy makers at different levels.For both types of informal learning, the links with formal processes play an important role to enable their contributions to transition.

6. Conclusion

First of all, this study contributes to a better understanding of the development of the flood management regime in the Dongting Lake area from 1949 to 2009. Our findings indicate that both double-loop and triple-loop learning toward IFM have taken place. It implies that transition toward the IFM regime has started in Dongting Lake area. However, there is evidence to show that triple-loop learning toward IFM has still been constrained by the incompatibilities between what is on paper and the real implementation. In addition, we found that, despite the existence of cross-sectoral collaboration at national level, decision-making and implementation processes at provincial level

[22] CCICED refers to China Council for International Cooperation on Environment and Development and is a think tank of the central government.

have not fully developed to support triple-loop learning of the regime structure change toward IFM and need to be improved. For example, given the powerful role of the provincial government in encouraging cross-sectoral collaboration, we suggest formal collaboration mechanisms be developed, where the provincial government acts as a facilitator. Such formal mechanisms should be complemented by informal learning processes where different departments recognize their interdependencies, deal with their differences constructively, and build up trust with each other (Pahl-Wostl, 2006). Meanwhile, the river basin commission's power over the enforcement of flood management strategic plan of Dongting Lake should be reinforced in order to avoid flood risk transfer to other regions in the Yangtze River basin. Besides, an increasing involvement of non-governmental stakeholders and experts from other disciplines, such as ecologists and political scientists, in strategic planning is essential for triple-loop learning toward IFM. In this way, strategic planning can be based on a better understanding of the ecological and socio-economic implications of flood management measures. The transition toward an ecological approach for water management in the Netherlands illustrates the importance of the engagement of ecologists for the transition (Van der Brugge, 2009).

It can be expected that, in China, triple-loop learning in Dongting Lake flood management was mostly initiated by the central government, given the hierarchical setting and the importance the central government has long attached to flood control in China. Top-down driven learning can sometimes be very effective, for example, the rapid implementation of floodplain restoration directly after the 1998 flood. The central government, who was inspired by the US experiences, ordered large-scale implementation. This is quite different to what happened in, for example, the Netherlands, where the ecological approach has become mainstream through a series of learning processes initiated by government agencies and researchers. However, top-down driven learning can also easily block transition toward IFM. For instance, the top level's preference for investing in structural measures could further reinforce the mindset of local water managers in favor of the traditional flood control regime. Local policy-makers may prefer structural solutions over non-structural measures, because the former are more visible to the central government (GHD Austria and IWHR, 2009).

Additionally, we also discussed two types of informal learning processes, including policy experiments and pilot schemes initiated by non-government actors, and their contributions to the transition. Such contributions were enabled by their links with formal policy processes. In fact, the first type, i.e. policy experiment, has long been employed by the Chinese government in different sectors. Rather than perceiving policy experiments to be risky or threatening their position by politicians in some Western countries, the Chinese government regards policy experimentation as a way to explore complexity and uncertainty in the fluid context where new policies are to be implemented (Xia and Pahl-Wostl, 2012b).

In addition to the findings on flood management regime development above, our study has added a new analytical method to the current transition research, which links regime components and three different societal learning types, to investigate whether and to what extent regime transition has taken place. The study also provides a special case of social-ecological system in China. However, our findings challenge the assumption of the regime change always being a step-wise move through the phases of single to double and on to triple-loop learning. In the Dongting case, triple-loop learning of floodplain restoration rapidly took place without an explicit period of double-loop learning, mainly due to the urgency of the problem and the "efficient and effective" top-down decision-making structure in Chinese flood management. Besides, triple-loop learning in institutions and infrastructures does not necessarily start after triple-loop learning in management paradigm. In the Dongting case, the announcement of new paradigm by the powerful mainstreaming actor started after the triple-loop learning of, for example, land use and population control in the floodplain and floodplain restoration.

Finally, our findings show clearly that concepts for societal transitions and societal learning are to some extent context dependent. A strong central government, as is the case in China, seems to be very effective and efficient in initiating and promoting changes to a considerable degree. However, such a top-down approach does not seem to be sufficient to sustain changes toward a profound structural transition. Further long-term studies and comparative analyses of transitions in different cultural settings are needed.

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